

My Eye-Camera AI Research

BRSET vs mBRSET: a full guide in simple words

Research by Gaurav Poudel

Guide written on 28 September 2026

Made from the files in the `BRSETvsmBRSET` folder

How to read this guide

Every hard word is explained the first time it shows up. There is also a word list (a glossary) at the end. You can read it from start to finish, or jump to the part you need.

What is inside

1. The story in one page
2. Why this matters
3. Words you need to know
4. The big question
5. What other scientists already found
6. The two photo collections (datasets)
7. The plan: how the study works
8. How we will measure "fair" and "good"
9. What I have already done
10. Where things stand right now
11. What I will do next, step by step
12. Problems I already know about, and how to fix them
13. Honest limits of the study
14. Risks and how to handle them
15. What is inside my folder
16. The rules I must follow
17. Word list (glossary)
18. One-page checklist

1. The story in one page

Some people have a sickness called **diabetes**. Diabetes means there is too much sugar in the blood. Over many years, that extra sugar can hurt the tiny blood vessels at the back of the eye. This damage is called **diabetic retinopathy**. Scientists shorten it to **DR**. If nobody catches DR early, a person can slowly go blind.

The good news is that doctors can catch DR early. They take a photo of the back of the eye. Then they look at the photo for signs of damage. This is called **screening**.

There are two kinds of cameras for these photos:

- A **big clinic camera**. It sits on a table in a hospital. It is expensive, and it takes very clear photos.
- A **small handheld camera**. It clips onto a phone. It is cheap and easy to carry to villages and small towns. But its photos look different. They can be darker, blurrier, or a different color.

Today, computers can look at these eye photos and say "this eye looks sick" or "this eye looks healthy". The computer program that does this is called an **AI model**. Most of the best AI models learned from clear clinic photos.

So here is my research, in one sentence:

If an AI model learns from clinic photos, and then we use it on handheld photos, it gets worse. **Does it get worse by the same amount for everybody, or does it get much worse for some groups of people, like older people, women, or people with less schooling or no health insurance?**

If the AI fails more for some groups, that is **unfair**. And those groups are often the same people who need help the most.

2. Why this matters

Handheld cameras are the cheapest way to bring eye screening to poor areas. Health companies want to buy an AI model that someone else already built and plug it into their handheld camera. This is quicker and cheaper than building a new one.

But if that borrowed AI quietly misses sick eyes in older or poorer people, it could hurt exactly the people it was meant to help. Nobody would notice, because the total score might still look "okay".

My study checks this. The answer helps two groups:

- **Scientists**, because nobody has joined these two questions (camera change and fairness) before.
- **Companies and health workers**, because they need to know if they can trust a borrowed model, or if they must collect their own photos and retrain it first.

3. Words you need to know

Word	What it means in simple words
Diabetes	A sickness where there is too much sugar in the blood.
Diabetic retinopathy (DR)	Eye damage caused by diabetes. It can lead to blindness.
Retina / fundus photo	The retina is the back wall inside your eye. A fundus photo is a picture of it.
Screening	Checking lots of people to find the sick ones early.
AI model	A computer program that learned from examples, like a student who learned from flashcards.
Training	Teaching the AI by showing it thousands of photos with the right answers.
Testing	Giving the AI new photos it has never seen and checking its answers.
Dataset	A big collection of photos plus information about each one.
Foundation model	A very big AI that first learned from millions of photos in general. Later it gets a little extra training for one special job.
Fine-tuning	That little extra training for one special job.
Subgroup	A smaller group of patients, like "women" or "people over 50".
Fairness	The AI works equally well for every subgroup.
Domain shift	The new photos look different from the photos the AI learned from.

4. The big question

The main question is:

When an AI model moves from clinic cameras to handheld cameras, **does the drop in accuracy fall evenly on everyone**, or does it fall harder on some groups?

The groups I will check are:

- **Age:** younger people vs older people.
- **Sex:** men vs women.
- **Education:** people who can read vs people who cannot read.
- **Insurance:** people with health insurance vs people without it.

There is also a second question:

How much of the drop is because of the **camera**, and how much is because the **patients** are different people from a different place?

This second question is tricky. I explain why in part 13.

5. What other scientists already found

Before starting, I read **39 research papers**. This is called a **literature review**. It helped me see what is already known, so I do not repeat someone else's work.

The most important paper: Li et al. 2026

A team in Denmark (Li and others) did almost my first idea already. They taught AI models with clinic photos (BRSET) and tested them on handheld photos (mBRSET). The score went down a lot. On clinic photos the models scored between 0.90 and 0.98. On handheld photos they scored between 0.70 and 0.85. (The score is explained in part 8. Higher is better and 1.0 is perfect.)

But they did not check fairness at all. They never asked if some groups lost more than others. That missing part is exactly where my research lives.

Their paper is a **preprint**. That means it is shared online but other scientists have not officially checked it yet.

The second important paper: Fernandes et al. 2025

This team checked fairness on the handheld photos (mBRSET) by age, sex, education and insurance. They found a gap of about 10 points between age groups for one model. **But they only used handheld photos.** They did not check what happens when the camera changes.

So what is new in my work?

Think of two puzzle pieces:

- Piece 1: "The score drops when the camera changes." (Li did this.)
- Piece 2: "Some groups get lower scores on handheld photos." (Fernandes did this.)

Nobody has put the two pieces together. My study asks: when the score drops because of the camera, does the drop hit some groups harder? I also add a clever camera check using two clinic cameras (part 7). No paper I found does that.

Other useful things I learned

- Special eye AI models do **not always** beat general AI models. In Li's study, a general model called DINOv3 beat the eye models.
- Models trained on medical photos can score higher but be **less fair**, according to one study (Khan et al. 2023).

- Some studies found **no** fairness gap. So "there is no gap" is a real possible answer for me too. That is okay. It is still a real result.
- Handheld photos often make the AI miss **mild** disease more than severe disease. So I will also check results by how sick the eye is.

6. The two photo collections (datasets)

Both datasets come from Brazil. They are on a website called **PhysioNet**, which shares medical data with approved researchers.

	BRSET	mBRSET
Camera	Big clinic cameras (Canon and Nikon)	Small handheld camera (Phelcom Eyer, clips onto a phone)
Place	3 eye clinics in São Paulo	A community diabetes event in Itabuna, Bahia
Number of photos	16,266	5,164
Number of patients	8,524	1,291
People with diabetes	About 16%	About 97%
People with DR	About 7%	About 23%
Info about each person	Age, sex, camera	Age, sex, education, insurance

A few important facts:

- **No person is in both datasets.** They are different people from different cities.
- **Education and insurance are only in mBRSET.** So for those two groups I can only look at handheld results.
- **About 92% of mBRSET people have no insurance.** Only around 100 people have insurance. That is a small group, so results for it will be shaky.
- **There is no race information** in either dataset, so I cannot study race.
- **BRSET has two cameras inside it:** Canon (about 65% of photos) and Nikon (about 35%). This is very useful, as part 7 explains.

7. The plan: how the study works

The AI models I will use

1. **RETFound**. A big AI made just for eye photos. It learned from 1.6 million eye photos.
2. **DINOv3**. A big general AI that learned from ordinary photos (cats, cars, trees). It did best in Li's study.
3. **VisionFM** (maybe). Another eye AI. Only if there is time.

All of them are the "ViT-Large" size. That is a big model type, so it needs a good graphics card. My computer has an **RTX 4060 with 8 GB** of memory.

The simple job the AI does

The AI answers one yes-or-no question for each photo:

- **No** = the eye has no DR (grade 0).
- **Yes** = the eye has some DR (grades 1 to 4).

(Doctors grade DR from 0 to 4 using a scale called **ICDR**. I join grades 1 to 4 into "yes".)

Fair splitting of the photos

Each person can have several photos. If one person's photos went into both the "learning" pile and the "test" pile, the AI could cheat by remembering that person. So I split by **person**, not by photo. I also use the **same splits that Li et al. used**, so my results can be compared with theirs.

I only use BRSET people **with diabetes**, so BRSET looks more like mBRSET.

The experiments

Experiment 1: Clinic test. Teach the AI on BRSET clinic photos. Test it on other BRSET clinic photos. This is the "home game". It shows how good the AI is when nothing changes.

Experiment 2: Handheld test (the main one). Take that same AI, with **no extra training**, and test it on mBRSET handheld photos. This is the "away game". It copies real life, where a company borrows a model and plugs it in.

Experiment 3 (optional): Fine-tuning. Give the AI a little extra training on some handheld photos. Then check if the score and the fairness get better.

Camera check: Canon vs Nikon. Inside BRSET, the patients are from the same clinics, but the photos come from two different cameras. If I teach on Canon and test on Nikon, any drop is mostly due to the **camera alone**. This helps me guess how much of the BRSET to mBRSET drop is about the camera, and how much is about the different patients.

Sickness-level check. I will also check the results for each DR grade separately, because mild disease is often missed more on handheld photos.

The training settings

These are the "recipe" settings, fixed in advance so I cannot change them to get nicer results:

- Photo size: 224 × 224 pixels.
- Learning method: Adam, learning speed 0.00001.
- 16 photos at a time.
- Up to 50 rounds of learning, but stop early if the AI stops getting better (after 8 rounds with no improvement).
- A fixed "random seed" of 42, so anyone can repeat my run and get the same result.

8. How we will measure "fair" and "good"

AUROC: the main score

AUROC is a score from 0.5 to 1.0 that tells how well the AI can tell sick eyes from healthy eyes.

- **1.0** = perfect. It always gets it right.
- **0.5** = no better than flipping a coin.
- A drop of **0.10** is a big deal.

The gap

For each group (like men vs women), I find the AUROC of each side. The **gap** is the difference. Example: if women score 0.85 and men score 0.80, the gap is 0.05. A bigger gap means less fair. (These numbers are just an example, not my results.)

Change in gap: the key new number

I measure the gap on clinic photos, then again on handheld photos. Then I look at how much the gap **grew or shrank**. If the gap grows when we switch cameras, the camera change hurts some people more than others. This is the heart of my study.

Equity-scaled AUC

This is one number that mixes "how good" with "how fair". It goes down if the groups are unequal. It makes results easier to compare.

Confidence intervals (how sure am I?)

With small groups, results can happen by luck. So I use a method called **bootstrap**. The computer re-picks the patients randomly **1,000 times** and redoes the math each time. That gives a range, like "between 0.78 and 0.84". If the range is very wide, I am not very sure, and I will say so.

Calibration (ECE)

Sometimes the AI says "I am 90% sure this eye is sick" when it is really right far less often. **Calibration** checks whether the AI's confidence is honest. ECE is the number I use for it. Li found that all models were over-confident.

9. What I have already done

Here is everything done so far, in order.

When	What I did
1 Sep 2026	Read and wrote notes on 39 papers in five topics. Wrote a summary (synthesis), a list of papers (index), a reference file, and a short research introduction.
1 Sep 2026	Found Li et al. 2026 and changed my research question to focus on fairness , because the plain camera question was already answered.
2 Sep 2026	Wrote and locked the study plan (the study protocol). Locking it means I cannot change the rules later to get nicer results.
Early Sep	Checked my plan against Li et al.'s exact method, so my results can be compared with theirs. Got their patient splits.
Early Sep	Wrote all the computer code: loading photos, building the models, training, testing, fairness math and bootstrap. Also wrote a small automatic test.
2 Sep 2026	Set up Python with GPU (graphics card) support and checked it works.
Early Sep	Ran a tiny " plumbing test " with 40 practice photos, just to check that every part of the code runs from start to end.
Early Sep	Ran a bigger practice study using free public photo sets: APTOS (from India) for learning and MESSIDOR-2 (from France) for testing.
Early Sep	Wrote first drafts of the thesis chapters: methods, related work, results (empty for now), and discussion with limits.
Early Sep	Re-checked that nobody had published my exact idea yet. Nobody has.
8 Sep 2026	Saved everything in Git (a tool that keeps a history of your files).
By 28 Sep 2026	PhysioNet approved my access. I signed the data agreements for both BRSET and mBRSET.
28 Sep 2026	Started downloading both datasets to <code>C:\data\brset_mbrset</code> , using a fast downloader that grabs many files at once.

What the practice runs showed

The plumbing test used only 40 learning photos and 24 test photos. Its scores were very low (AUROC about 0.33). That is expected and does not matter. Its only job was to prove the code runs. **These numbers must never go in the thesis.**

The practice study trained a smaller ViT model on 800 APTOS photos for 10 rounds. It learned well. Its check score (validation AUROC) climbed from 0.90 in round 1 to its best of **0.994 in round 6**, then slowly slipped. Each round took about 37 seconds. But the final test on MESSIDOR **crashed**, because a bug marked every MESSIDOR photo as "healthy". You cannot measure "sick vs healthy" if there are no sick photos. This bug is listed in part 12.

Important: none of these practice results are real thesis results. The thesis results chapter is still empty, and it should stay empty until the real experiments run.

10. Where things stand right now

-  Literature review: **done**.
-  Study plan: **done and locked**.
-  Code: **written** and tested on practice data.
-  Thesis drafts: methods, related work and discussion are **drafted**. Results are **waiting**.
-  PhysioNet access: **approved**, agreements signed.
-  Download: **in progress**. At about 7:30 PM on 28 September, mBRSET had about 2,151 of 5,166 files, and BRSET had about 2,198 of 16,268 files. mBRSET should finish around midnight. BRSET should finish around midday on 29 September. Some files may fail the first time. Running the same download command again fixes them.
-  Real experiments: **not started yet**. They need the download to finish first.

11. What I will do next, step by step

1. **Finish the download.** Run the download command again at the end to catch any failed files.
2. **Check the files.** Count them. There should be about 16,266 BRSET photos and 5,164 mBRSET photos. The download tool also checks each file with a code called SHA256, to make sure nothing is broken.
3. **Link the data to the project folder.** Make a shortcut (a "junction") from `C:\data\brset_mbrset` into the project, so the code can find the photos. The photos stay outside OneDrive, so they do not upload to the cloud.
4. **Check the labels and IDs** against Li et al.'s splits, to make sure the same people are in the same piles.
5. **Fix the photo loader.** The code needs to know that BRSET and mBRSET keep their photos in different folders, and that their information columns have different names.
6. **Fix the MESSIDOR bug** in the practice code, and make the fairness math handle a group with no sick eyes without crashing.
7. **Download the model weights** for RETFound and DINOv3. (Weights are the "brain" files the model already learned.)
8. **Run Experiment 1** (clinic test) with RETFound, then DINOv3.
9. **Run Experiment 2** (handheld test) with the same models, no extra training.
10. **Run the Canon vs Nikon camera check.**
11. **Do the fairness math:** subgroup scores, gaps, change in gap, equity-scaled AUC, bootstrap ranges, calibration, and the sickness-level breakdown.
12. **Compare my overall numbers with Li et al.** If mine are close to theirs, that shows my setup is right.
13. **Maybe run Experiment 3** (fine-tuning) and VisionFM, if there is time.
14. **Make the charts and tables**, and write the results chapter.
15. **Update the discussion** with what the results mean.
16. **Re-check the novelty** one more time, right before submitting, to make sure nobody published the same idea.
17. **Submit** the thesis, and maybe a paper.

12. Problems I already know about, and how to fix them

Problem	Why it matters	How to fix it
Every MESSIDOR photo was marked "healthy" in the practice study.	You cannot score sick vs healthy with no sick photos, so the math breaks and then crashes.	Fix how MESSIDOR grades are read. Also make the fairness code skip a group that has only one kind of answer, instead of crashing.
The loader expects one photo folder for everything.	BRSET and mBRSET keep photos in different folders.	Give each dataset its own photo folder in the settings file.
mBRSET uses different column names.	The code looks for BRSET names, like <code>image_id</code> and <code>DR_ICDR</code> . mBRSET uses names like <code>file</code> , <code>final_icdr</code> , <code>patient</code> , <code>age</code> , <code>sex</code> , <code>educational_level</code> and <code>insurance</code> .	Add a small "translation" step that renames mBRSET columns to match.
The download is slow.	It could take until 29 September.	Let it run. Rerun the same command to retry any failed files.
The model is big for an 8 GB graphics card.	It could run out of memory.	Keep 16 photos per step. If needed, use tricks like mixed precision or gradient accumulation.

13. Honest limits of the study

Good science is honest about what it **cannot** prove. Here are my limits:

- **Two things change at once.** BRSET and mBRSET differ in camera, city, patients, how sick people are, and who graded the photos. So a drop cannot be blamed only on the camera. The Canon vs Nikon check helps, but it does not solve this fully.
- **Education and insurance only exist in mBRSET.** So I can measure the "change in gap" only for age and sex. For education and insurance, I can only show the gap on handheld photos.
- **The insured group is tiny** (about 100 people). Its results are **exploratory**, which means "interesting, but not strong proof".
- **No race data.** I cannot say anything about race or skin color.
- **Brazil only.** The results may not be the same in Nepal, India or other countries.
- **Only a few models.** Other models might behave differently.
- **"No gap" is possible.** If there is no gap, that is still a real and useful finding.

14. Risks and how to handle them

Risk	What I can do
Someone publishes my idea first. The team that made BRSET and mBRSET (the Nakayama and Celi group) is very active in this area.	Work fast. Re-check the latest papers before submitting.
Li et al. adds fairness in their final paper.	Keep watching their paper. My Canon vs Nikon check and "change in gap" would still add something new.
mBRSET version 2 comes out , maybe with race data.	Check for it. It could open a new part of the study.
Results are too uncertain because groups are small.	Report the ranges honestly and mark small groups clearly.
Computer problems (download or memory).	Keep data outside OneDrive, rerun failed downloads, and use memory-saving settings.

15. What is inside my folder

The main folder is `C:\Users\gaura\OneDrive\Desktop\BRSETvsmBRSET`.

Folder or file	What is in it
<code>study-protocol.md</code>	The locked study plan. The rulebook for the whole study.
<code>literature-review/</code>	The 39 paper notes, the summary (<code>synthesis.md</code>), the list (<code>index.md</code>), references, and the short research introduction.
<code>thesis/</code>	Draft chapters: methods, related work, results (empty for now), and discussion with limitations.
<code>docs/</code>	Helper guides: the practice study, the access checklist, the Li et al. comparison, the novelty re-check, computer setup, and how to get practice data.
<code>configs/</code>	Settings files. One for the real study (<code>brset_mbrset.yaml</code>), and others for the practice runs.
<code>src/</code>	The main code: loading data, building models, training, testing, fairness groups and bootstrap.
<code>scripts/</code>	Small programs to download weights, prepare practice data, run an experiment, and check the computer setup.
<code>tests/</code>	An automatic test that checks the subgroup code.
<code>results/</code>	Output from the practice runs only. Not for the thesis.
<code>prompts/</code>	A log of how the literature review was made, plus ready-made instructions for future re-checks.
<code>C:\data\brset_mbrset</code>	Where the real photos are downloading. Kept outside OneDrive on purpose.

16. The rules I must follow

When I got access to the data, I signed an agreement called a **DUA** (Data Use Agreement). It is like a promise. The main rules are:

- Use the data only for this research.
- Do not share the photos or patient information with anyone who is not approved.
- **Do not upload patient data to AI chat tools** (like ChatGPT or other assistants). It is fine to share code, documents, and summary numbers (like "the AUROC was 0.85"), but never the photos or the patient tables.
- Do not try to find out who any patient is.
- Cite the datasets properly in the thesis.

17. Word list (glossary)

Accuracy drop

How much worse the AI does on new kinds of photos.

AUROC (or AUC)

A score from 0.5 (coin flip) to 1.0 (perfect) for telling sick from healthy.

Bootstrap

Re-picking patients at random many times to see how sure a result is.

Calibration / ECE

Whether the AI's "how sure I am" number is honest.

Confidence interval

A range where the true answer probably is, like "0.78 to 0.84".

Confound

When two things change together, so you cannot tell which one caused the result.

Dataset

A big collection of photos with information about each one.

DINOv3

A big general AI model that learned from everyday photos.

DR (diabetic retinopathy)

Eye damage caused by diabetes.

DUA

Data Use Agreement, the promise you sign to use the data safely.

Early stopping

Stopping training when the AI stops getting better, so it does not over-learn.

Epoch

One full round where the AI sees every training photo once.

Equity-scaled AUC

One number that mixes how good and how fair the AI is.

Exploratory

Interesting, but not strong enough to be proof.

Fine-tuning

A little extra training for a special job.

Foundation model

A huge AI trained on millions of images, later adapted to smaller jobs.

Fundus photo

A photo of the back of the eye.

Gap

The difference in score between two groups.

ICDR

The 0 to 4 scale doctors use to grade how bad DR is.

Literature review

Reading many papers to learn what is already known.

Novelty

The part of the work that is new and has not been done before.

PhysioNet

The website that shares these medical datasets with approved researchers.

Preprint

A paper shared online before other experts officially check it.

Protocol

The written plan of the study, fixed before starting.

RETFound

A big AI model made for eye photos.

Seed

A fixed number that makes "random" steps repeat the same way each time.

Subgroup

A smaller group of patients, like "women" or "people over 50".

ViT

Vision Transformer, a type of AI model that looks at a picture in small squares.

Weights

The saved "brain" of a model, everything it has learned.

18. One-page checklist

Done

- ☒ Read 39 papers and wrote the literature review
- ☒ Changed the question to focus on fairness
- ☒ Wrote and locked the study plan
- ☒ Matched the plan to Li et al.'s method and splits
- ☒ Wrote all the code and tested it on practice data
- ☒ Drafted methods, related work, and discussion chapters
- ☒ Got PhysioNet access and signed the agreements
- ☒ Started downloading both datasets

To do

- ☐ Finish and check the download
- ☐ Link the data folder into the project
- ☐ Fix the photo loader and the mBRSET column names
- ☐ Fix the MESSIDOR bug and the crash in the fairness math
- ☐ Download RETFound and DINOv3 weights
- ☐ Run Experiment 1 (clinic test)
- ☐ Run Experiment 2 (handheld test)
- ☐ Run the Canon vs Nikon camera check
- ☐ Do all fairness math and the sickness-level breakdown
- ☐ Compare with Li et al.
- ☐ (Optional) Experiment 3 and VisionFM
- ☐ Make charts and write the results chapter
- ☐ Re-check novelty
- ☐ Submit

This guide only describes plans and practice runs. It contains no real results from BRSET or mBRSET yet, because those experiments have not been run.